



# Girraween High School

## 2025

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

## Mathematics Extension 2

### General Instructions

- Reading time: 10 minutes
- Working time: 3 Hours
- Write using a black pen
- NESA approved calculators may be used
- Laminated reference sheets are provided
- Answer multiple-choice questions by completely colouring in the appropriate circle on your multiple-choice answer sheet.
- Answer questions in the writing booklet provided and show all relevant mathematical reasoning and/or calculations.

**Total Marks: 100**

### Section 1 (Pages 1 – 5) **10 Marks**

- Attempt Q1 - Q10
- Allow about 15 minutes for this section

### Section 2 (Pages 6-10) **90 marks**

- Attempt Q11 - Q16
- Allow about 2 hours and 45 minutes for this section

## Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

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### Question 1

The angle between the vectors  $\begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix}$  and  $\begin{pmatrix} -3 \\ 4 \\ 1 \end{pmatrix}$  is

- A)  $27^\circ$
- B)  $68^\circ$
- C)  $112^\circ$
- D)  $133^\circ$

### Question 2

The statement “For all positive integers  $n$ , there exists a positive integer  $p$  such that  $p = n^2$ ” is represented in mathematical symbols by:

- A)  $\forall n \in \mathbb{Z}^+ \exists p \in \mathbb{Z}^+ \text{ such that } p = n^2$
- B)  $\exists n \in \mathbb{Z} \exists p \in \mathbb{Z} \text{ such that } p = n^2$
- C)  $\forall n \in \mathbb{Z}^+ \exists p \in \mathbb{Z}^+ \text{ such that } p = n^2$
- D)  $\forall n \in \mathbb{Z} \exists p \in \mathbb{Z} \text{ such that } p = n^2$

*Examination continues next page...*

### Question 3

The contrapositive of “If the sum of the digits of an integer is a multiple of 3, the integer is a multiple of 3” is

- A) “If an integer is a multiple of 3 the sum of its digits will be a multiple of 3”.
- B) “If an integer is not a multiple of 3, the sum of its digits will NOT be a multiple of 3”.
- C) “If the sum of the digits of an integer is NOT a multiple of 3, the number will NOT be a multiple of 3”.
- D) “If the sum of the digits of an integer is NOT a multiple of 3, the integer is a multiple of 3”.

### Question 4

The converse of “If the sum of the digits of an integer is a multiple of 3, the integer is a multiple of 3” is

- A) “If an integer is a multiple of 3 the sum of its digits will be a multiple of 3”.
- B) “If an integer is not a multiple of 3, the sum of its digits will NOT be a multiple of 3”.
- C) “If the sum of the digits of an integer is NOT a multiple of 3, the number will NOT be a multiple of 3”.
- D) “If the sum of the digits of an integer is NOT a multiple of 3, the integer is a multiple of 3”.

*Examination continues next page...*

**Question 5**

$$\int \frac{6}{(2x-1)(x+1)} dx =$$

A)  $\ln \left| \frac{x+1}{2x-1} \right| + C$

B)  $\ln \left| \frac{2x-1}{x+1} \right| + C$

C)  $2 \ln \left| \frac{x+1}{2x-1} \right| + C$

D)  $2 \ln \left| \frac{2x-1}{x+1} \right| + C$

**Question 6**

$$i^{2025} =$$

A)  $i$

B)  $-i$

C)  $1$

D)  $-1$

*Examination continues next page...*

**Question 7**

It is given for the complex number  $z$  that  $|z - 2 - i| = 1$ . The maximum modulus for  $z$  is

- A) 1
- B)  $\sqrt{5}$
- C)  $\sqrt{5} - 1$
- D)  $\sqrt{5} + 1$

**Question 8**

Two of the roots of a monic polynomial of degree 3 with all real coefficients are 4 and  $2 + i$ .

The polynomial is

- A)  $z^3 - 6z^2 + 18z - 40$
- B)  $z^3 + 8z^2 - 21z + 20$
- C)  $z^3 - 8z^2 + 21z - 20$
- D)  $z^3 + 6z^2 + 18z - 40$

**Question 9**

A particle is moving with simple harmonic motion with a period of 12 seconds and an amplitude of 5 metres. If it starts at the centre of motion and is moving to the right, when will it first be 5 metres to the left of the centre of motion?

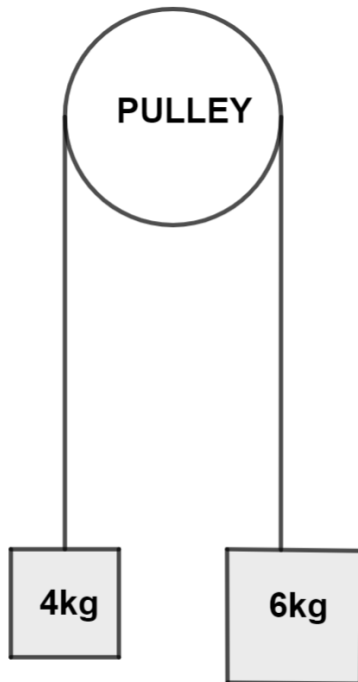
- A) After 3 seconds
- B) After 6 seconds
- C) After 9 seconds
- D) After 12 seconds

*Examination continues next page...*

**Question 10**

Two weights of 4kg and 6 kg respectively are attached to a pulley by the same string (see diagram).

The acceleration due to gravity is  $g$  metres per second squared.



The acceleration of the 6kg mass is

- A)  $10g \text{ m/s}^2$
- B)  $5g \text{ m/s}^2$
- C)  $\frac{g}{10} \text{ m/s}^2$
- D)  $\frac{g}{5} \text{ m/s}^2$

*Examination continues next page...*

## Section II

90 marks

Attempt Questions 11 - 16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

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### Question 11 (15 marks) Start a new writing page

a)  $z$  and  $w$  are the numbers  $1 - i$  and  $1 - i\sqrt{3}$  respectively.

(i) Show that  $\frac{z}{w} = \frac{(1+\sqrt{3})+i(\sqrt{3}-1)}{4}$  1

(ii) Express both  $z$  and  $w$  in exponential form 2

(iii) Hence show that  $\cos \frac{\pi}{12} = \frac{\sqrt{2}+\sqrt{6}}{4}$  2

b) (i) By letting  $(x + iy)^2 = -5 - 12i$ ,  $x, y$  real find the two square roots of  $-5 - 12i$  2

(ii) Hence solve the equation  $z^2 + (2 + i)z + (2 + 4i) = 0$  2

c) Shade the region on an Argand diagram where  $|z + 2 - 2i| \geq 2$  and 4

$$\frac{\pi}{2} \leq \text{Arg } z < \frac{3\pi}{4}$$

d) Find the six sixth roots of  $64i$ . You may leave your answer in modulus argument form. 2

*Examination continues next page...*

**Question 12** (15 marks) **Start a new writing page**

a) (i) Find the values of  $A, B$  and  $C$  if  $\frac{A}{(x+1)^2} + \frac{B}{(x+1)} + \frac{C}{(x-3)} = \frac{-13x-25}{(x+1)^2(x-3)}$  3

(ii) Hence find  $\int \frac{-13x-25}{(x+1)^2(x-3)} dx$  2

b) Find  $\int_1^4 \frac{1}{2+\sqrt{x}} dx$  4

c) Find  $\int (\ln(x))^2 dx$  3

d) Use the substitution  $t = \tan\left(\frac{x}{2}\right)$  to find  $\int \frac{1}{2+\cos x} \cdot dx$  3

*Examination continues next page...*

**Question 13** (15 marks) **Start a new writing page**

- a) A particle moves so that its displacement at time  $t$  seconds is given by

$$x = 4 \cos \frac{t}{2} - 4\sqrt{3} \sin \frac{t}{2}$$

- (i) Show that the particle is moving in simple harmonic motion. 1

- (ii) By expressing  $x$  in the form  $A \cos \left( \frac{t}{2} + \alpha \right)$  or otherwise, find the period and amplitude of the motion. 2

- b) A particle moves in simple harmonic motion so that its acceleration at time  $t$  is given by  $\ddot{x} = -9(x + 2)$ . If the particle passes through  $x = 1$  with a speed of  $12\text{m/s}$ , show that  $v^2 = 189 - 9x^2 - 36x$  and state the period, amplitude and centre of motion. 3

- c) A car with a mass of  $1000\text{kg}$  starts from rest at a traffic light and drives on to a horizontal freeway. The car's engine provides a total force of  $16000$  Newtons and the combined resistance from friction from the freeway's surface and air resistance is  $10v^2$  Newtons.

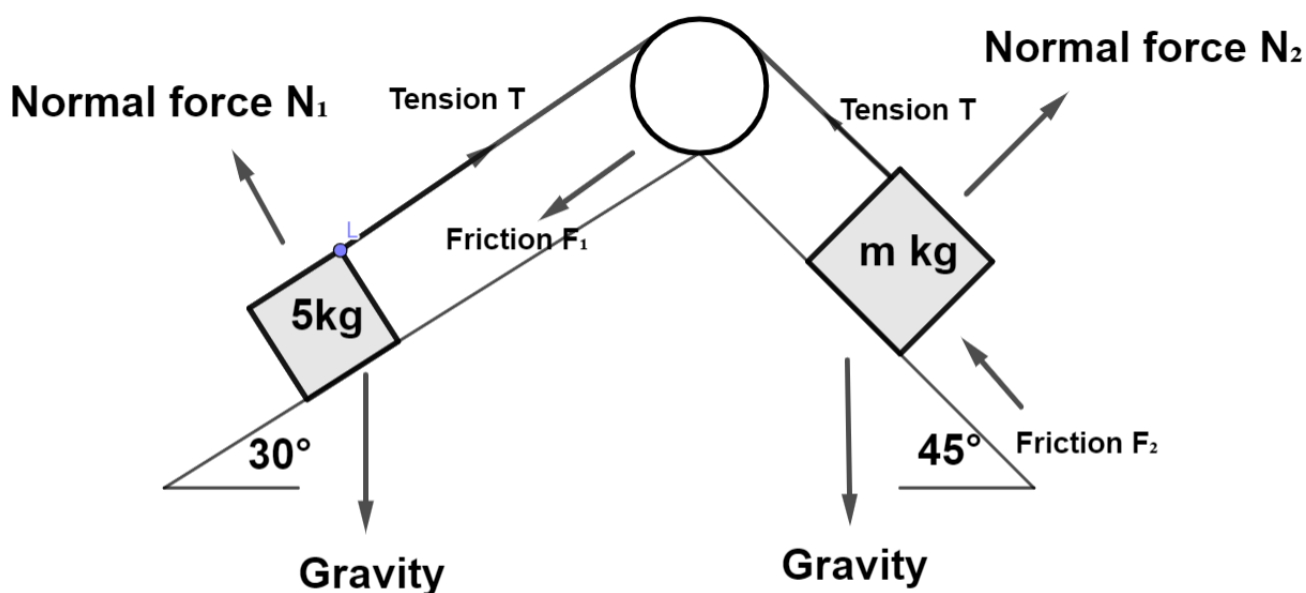
- (i) Show that the car's acceleration is given by  $\ddot{x} = 16 - 0.01v^2$  and find the speed the car can not exceed on the freeway (i.e. the speed it can never reach). 2

- (ii) Show that  $x = 50 \ln \left( \frac{1600}{1600 - v^2} \right)$  and find the distance the car has gone when it achieves a speed of  $30\text{m/s}$ . 3

*Examination continues next page...*

**Question 13 (continued)**

- d) Two square weights, one with mass  $5\text{ kg}$  and one with mass  $m\text{ kg}$  where  $m > 5$  are attached by a single rope running over a smooth pulley. The  $5\text{ kg}$  mass is on a slope of  $30^\circ$  to the horizontal and the  $m\text{ kg}$  mass is on a slope of  $45^\circ$  to the horizontal. The two weights are stationary as they are being held in place by friction but are on the point of slipping with the  $m\text{ kg}$  weight being about to fall down its slope, resulting in the  $5\text{ kg}$  weight being pulled UP its slope in response. Apart from friction, the weights are being held in place by gravity (with force of 10 times each object's mass) and by normal forces pushing out from the plane. Friction in each case is 0.6 times the normal force (note that friction and the normal force are different at each object but the tension in the rope is the same) (see diagram).



- (i) By resolving forces both parallel and perpendicular to the plane at the  $5\text{ kg}$  weight, 2  
show that the tension in the rope is equal to  $(15\sqrt{3} + 25)$  Newtons (you may treat tension as parallel to the plane).
- (ii) By resolving forces both parallel and perpendicular to the plane at the  $m\text{ kg}$  weight, 2  
find the mass ( $m$ ) of the second weight in kilograms.

*Examination continues next page...*

**Question 14** (15 marks) **Start a new writing page**

a) If  $I_N = \int_0^{\frac{\pi}{2}} \sin^N x \, dx$

(i) Show that  $I_N = \frac{N-1}{N} I_{N-2}$  2

(ii) Find the exact value of  $I_6$  2

b) (i) State the centre and radius of the sphere  $S$   $(x-2)^2 + (y+1)^2 + (z-4)^2 = 14$  2

(ii) By finding the perpendicular distance from the line  $L_1 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 9 \\ 6 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix}$  2

to the centre of the sphere  $S$ , show that the line  $L_1$  intersects with the circle in two places.

(iii) Find the two points of intersection of the line  $L$  and the sphere  $S$ . 3

(iv) Show that the line  $L_1$  is perpendicular to the line  $L_2 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}$  1

(v) Find the point of intersection of the lines  $L_1$  and  $L_2$  3

*Examination continues next page...*

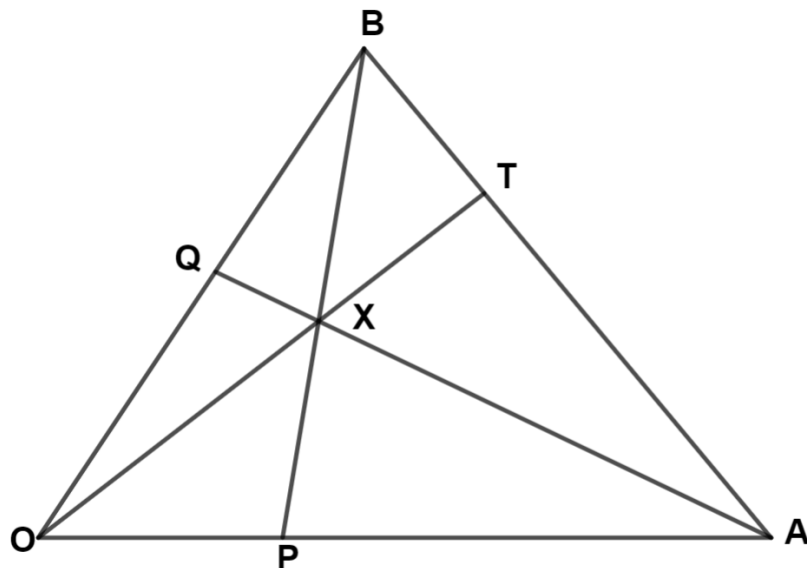
**Question 15** (15 marks) **Start a new writing page**

- a) Prove  $n! > 3^n$  for all positive integers  $n \geq 8$  using the method of mathematical induction. 3
- b) (i) Prove for all real numbers  $a$  and  $b$  that  $a^2 + b^2 \geq 2ab$  1
- 2
- (ii) Hence prove  $a^2 + b^2 + c^2 \geq ab + ac + bc$  for all real numbers  $a, b, c$ . 1
- (iii) Hence prove  $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \geq \frac{a+b+c}{abc}$  for all real numbers  $a, b, c$ . 1
- c) Prove by contradiction that  $\sqrt{29}$  is irrational. 2

*Question 15 continues next page...*

**Question 15 (continued)**

- d) In triangle  $OAB$ , the vector  $\overrightarrow{OA} = \underline{\underline{a}}$  and the vector  $\overrightarrow{OB} = \underline{\underline{b}}$ .  $P$  is on  $OA$  so that  $\overrightarrow{OP} = \frac{1}{3}\overrightarrow{OA}$  and  $Q$  is on  $OB$  so that  $\overrightarrow{OQ} = \frac{1}{2}\overrightarrow{OB}$ .  $X$  is at the intersection of  $AQ$  and  $BP$ .  $\overrightarrow{PX} = \lambda \overrightarrow{PB}$  and  $\overrightarrow{QX} = \mu \overrightarrow{QA}$ .  $T$  is on  $BA$  so that  $O, X$  and  $T$  are collinear. (see diagram).



(i) Show that  $\overrightarrow{OX} = \frac{1}{3}(1 - \lambda) \underline{\underline{a}} + \lambda \underline{\underline{b}}$  2

(ii)  $\overrightarrow{OX}$  can also be shown to be  $\mu \underline{\underline{a}} + \frac{1}{2}(1 - \mu) \underline{\underline{b}}$  (Do NOT prove this!) 2

By finding  $\lambda$  and  $\mu$ , show that  $\overrightarrow{OX} = \frac{1}{5} \underline{\underline{a}} + \frac{2}{5} \underline{\underline{b}}$

(iii) Given that  $T$  is collinear with  $O$  and  $X$  and is on the line  $AB$ , express  $\overrightarrow{OT}$  in terms of  $\underline{\underline{a}}$  and  $\underline{\underline{b}}$ . 3

*Examination continues next page...*

**Question 16** (10 marks) **Start a new writing page**

- a) Sketch the curve  $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} t \cos t \\ t \sin t \end{pmatrix}$  from  $t = 0$  to  $t = 2\pi$  2
- b) (i) State the solutions to  $z^9 - 1 = 0$  (*leave your solutions in modulus/argument form*). 1
- (ii) Hence using that  $z^9 - 1 = (z^3 - 1)(z^6 + z^3 + 1)$ , state the solutions to  $z^6 + z^3 + 1 = 0$ . 1
- (iii) Show that if  $z = \cos \theta + i \sin \theta$ ,  $z^n + \frac{1}{z^n} = 2 \cos n\theta$ . 1
- (iv) Express  $z^6 + z^3 + 1$  in terms of real quadratic factors. 2
- (v) Hence show that 
$$\cos 3\theta + \frac{1}{2} = 4\left(\cos \theta - \cos \frac{2\pi}{9}\right)\left(\cos \theta - \cos \frac{4\pi}{9}\right)\left(\cos \theta + \cos \frac{\pi}{9}\right)$$
 2
- c) A projectile with mass  $m$  kg is launched vertically from the ground with a velocity of  $v_0$  metres per second where  $v_0 < 200$  metres per second. It experiences gravity of  $10m$  Newtons and air resistance of  $0.05mv$  Newtons against its direction of motion, where  $v$  is the current velocity. If the projectile lands in the same place 5 seconds after launch, find its initial velocity  $v_0$ . 6

***End of Examination.***



# 2025 Year 12 Trial

## Mathematics Extension 2

### Multiple Choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

**Sample**  $2 + 4 =$  (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐  
correct

- |     |   |                                  |   |                                  |   |                                  |   |                                  |
|-----|---|----------------------------------|---|----------------------------------|---|----------------------------------|---|----------------------------------|
| 1.  | A | <input type="radio"/>            | B | <input checked="" type="radio"/> | C | <input type="radio"/>            | D | <input type="radio"/>            |
| 2.  | A | <input type="radio"/>            | B | <input type="radio"/>            | C | <input checked="" type="radio"/> | D | <input type="radio"/>            |
| 3.  | A | <input type="radio"/>            | B | <input checked="" type="radio"/> | C | <input type="radio"/>            | D | <input type="radio"/>            |
| 4.  | A | <input checked="" type="radio"/> | B | <input type="radio"/>            | C | <input type="radio"/>            | D | <input type="radio"/>            |
| 5.  | A | <input type="radio"/>            | B | <input type="radio"/>            | C | <input type="radio"/>            | D | <input checked="" type="radio"/> |
| 6.  | A | <input checked="" type="radio"/> | B | <input type="radio"/>            | C | <input type="radio"/>            | D | <input type="radio"/>            |
| 7.  | A | <input type="radio"/>            | B | <input type="radio"/>            | C | <input type="radio"/>            | D | <input checked="" type="radio"/> |
| 8.  | A | <input type="radio"/>            | B | <input type="radio"/>            | C | <input checked="" type="radio"/> | D | <input type="radio"/>            |
| 9.  | A | <input type="radio"/>            | B | <input type="radio"/>            | C | <input checked="" type="radio"/> | D | <input type="radio"/>            |
| 10. | A | <input type="radio"/>            | B | <input type="radio"/>            | C | <input type="radio"/>            | D | <input checked="" type="radio"/> |

# Mathematics Assessment Solutions

Girraween High School

page /

☐ Year 11

☐ Extension 1

Year 2025

☒ Year 12

☒ Extension 2

Task Number 4 TRIAL

## Suggested solutions

## Marker's Feedback

Multiple Choice:

$$\begin{aligned} (1) \cos \theta &= \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| |\underline{b}|} \\ &= \frac{\begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 4 \\ 1 \end{pmatrix}}{\sqrt{14} \times \sqrt{26}} \\ &= \frac{7}{\sqrt{364}} \end{aligned}$$

$$\theta = 68^\circ \quad \textcircled{B}$$

$$(2) \forall n \in \mathbb{Z}^+ \Rightarrow p \in \mathbb{Z}^+ \text{ such that } p = n^2. \quad \textcircled{C}$$

(3) Contrapositive:

"If an integer is not a multiple of 3 its digits will not add to a multiple of 3."  $\textcircled{B}$

(4) Converse: "If an integer is a multiple of 3 its digits will add to a multiple of 3."  $\textcircled{A}$

$$(5) \int \frac{6}{(2x-1)(x+1)} \cdot dx$$

$$= \int \frac{4}{(2x-1)} - \frac{2}{(x+1)} \cdot dx$$

$$= 2 \ln|2x-1| - 2 \ln|x+1| + C$$

$$= 2 \ln \left| \frac{2x-1}{x+1} \right| + C \quad \textcircled{D}$$

$$\frac{A}{2x-1} + \frac{B}{x+1} = -\frac{6}{(2x-1)(x+1)}$$

$$A(x+1) + B(2x-1) = -6 \quad (1)$$

Sub.  $x = -1$  in (1):

$$-3B = 6 \Rightarrow B = -2.$$

Sub.  $x = \frac{1}{2}$  in (1):

$$\frac{3A}{2} = 6 \Rightarrow A = 4.$$

## Suggested solutions

## Marker's Feedback

Multiple Choice [continued]:

(6)  $i^{2025}$

$= i^{2024} \times i$

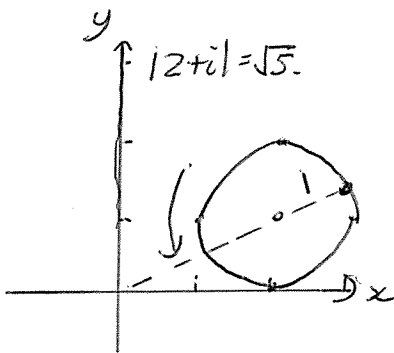
$= [i^4]^{506} \times i$

$= 1 \times i$

$= i$

(A)

(7)  $|z - 2 - i| = 1$



Maximum modulus

$= \sqrt{5} + 1$

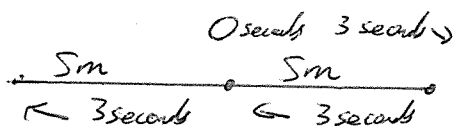
(D)

(8) By conjugate root theorem, other root is  $2 - i$ .

$$\begin{aligned} \therefore \text{Polynomial} &= (z - 4)(z - 2 - i)(z - 2 + i) \\ &= (z - 4)(z^2 - 4z + 5) \\ &= z^3 - 8z^2 + 21z - 20 \end{aligned}$$

(C)

(9)



Will be 5m LEFT after  
 $3 + 3 + 3 = 9$  seconds.

(C)

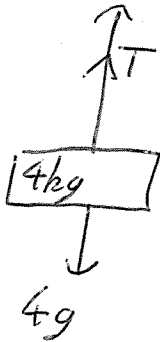
## Suggested solutions

## Marker's Feedback

Multiple Choice Continued):

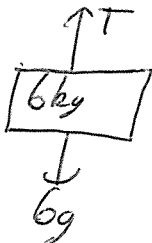
Q.(10)

At 4kg mass:



$$T - 4g = 4a \quad (1)$$

At 6kg mass:



$$6g - T = 6a \quad (2)$$

$$(1) + (2) \quad 2g = 10a$$

$$\underline{\underline{\frac{g}{5} = a}}$$

①

| Suggested solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marker's Feedback |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <p>Q.11)(a)</p> <p>(i) <math>\frac{z}{w} = \frac{1-i}{1-i\sqrt{3}} \times \frac{(1+i\sqrt{3})}{(1+i\sqrt{3})}</math> <u>1</u> to here.<br/> <math>= \frac{1+i\sqrt{3}-i+\sqrt{3}}{4}</math> Well done.<br/> <math>= \frac{(1+\sqrt{3})+i(\sqrt{3}-1)}{4}</math></p> <p>(ii) <math>z = \sqrt{2} e^{-\frac{\pi i}{4}}</math> <u>2</u><br/> <math>w = 2 e^{-\frac{\pi i}{3}}</math></p> <p>(iii) <math>\frac{z}{w} = \frac{1}{\sqrt{2}} e^{\frac{\pi i}{12}}</math><br/> <math>= \frac{1}{\sqrt{2}} \left[ \cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right]</math> <u>1</u></p>                                                                                                                                                                                                                                               |                   |
| <p>Equating real parts with (i):</p> $\frac{1}{\sqrt{2}} \cos \frac{\pi}{12} = \frac{1+\sqrt{3}}{4} \times \sqrt{2}$ $\cos \frac{\pi}{12} = \frac{\sqrt{2}+\sqrt{6}}{4}$ <p>(b)(i) <math>(x+iy)^2 = -5-12i</math><br/> <math>(x^2-y^2)+2ixy = -5-12i</math></p> <p>Equating reals, <math>x^2-y^2 = -5</math> (1)<br/> Equating imaginaries, <math>2xy = -12</math><br/> <math>y = -\frac{6}{x}</math> (2) <u>1</u></p> <p>Sub. (2) in (1): <math>x^2 - \left(-\frac{6}{x}\right)^2 = -5</math><br/> <math>x^2 - \frac{36}{x^2} = -5</math><br/> <math>x^4 + 5x^2 - 36 = 0</math><br/> <math>(x^2-4)(x^2+9) = 0</math> <u>2</u></p> <p>As <math>x</math> is real, <math>x = \pm 2</math>. <u>1</u><br/> As <math>y = -\frac{6}{x}</math>, <math>y = \mp 3</math>.</p> <p><math>\therefore \sqrt{-5-12i} = \pm(2-3i)</math></p> |                   |

## Suggested solutions

## Marker's Feedback

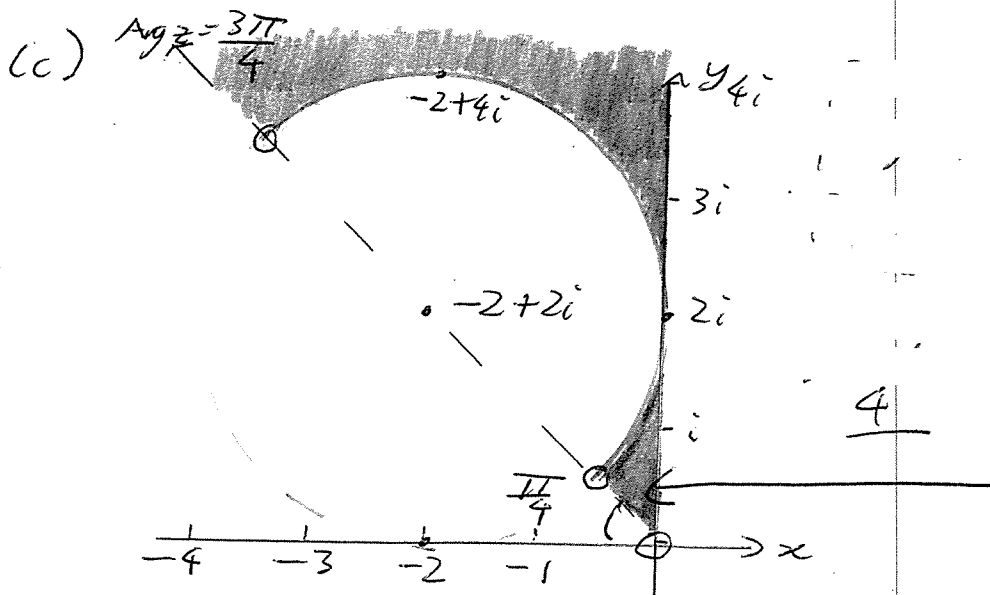
Q.11(b)(ii)  $z^2 + (2+i)z + (2+4i) = 0$

$$z = \frac{-2-i \pm \sqrt{(2+i)^2 - 4 \times 1 \times (2+4i)}}{2 \times 1}$$

$$= \frac{-2-i \pm \sqrt{-5-12i}}{2}$$

$$= \frac{-2-i \pm (2-3i)}{2}$$

$z = -2i$  or  $z = -2+i$



(d)  $\sqrt[6]{64i}$

$$= \sqrt[6]{64 \text{cis} \frac{\pi}{2}}$$

+1 for 1 answer  
+1 for all answers

$$= 2 \text{cis} \frac{\pi}{12}, 2 \text{cis} \frac{5\pi}{12}, 2 \text{cis} \frac{3\pi}{4},$$

$$2 \text{cis} \frac{13\pi}{12}, 2 \text{cis} \frac{17\pi}{12}, 2 \text{cis} \frac{7\pi}{4}.$$

| Suggested solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marker's Feedback                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <p>(12)(a) (i)</p> $\frac{A}{(x+1)^2} + \frac{B}{x+1} + \frac{C}{x-3} = \frac{-13x-25}{(x+1)^2(x-3)}$ <p><math>\therefore A(x-3) + B(x+1)(x-3) + C(x+1)^2 = -13x-25 \quad (1)</math></p> <p>Sub. <math>x = -1</math> in (1): <math>-4A = -13(-1) - 25 \Rightarrow \underline{A = 3}</math></p> <p>Sub. <math>x = 3</math> in (1): <math>C(3+1)^2 = -13 \times 3 - 25</math><br/> <math>16C = -64 \Rightarrow \underline{C = -4}</math></p> <p>Sub. <math>x = 0, A = 3, C = -4</math> in (1): <math>3 \times -3 - 3B + -4 = -25</math><br/> <math>-3B = -12 \Rightarrow \underline{B = 4}</math></p> <p><math>A = 3, B = 4, C = -4</math></p> <p>(ii) Hence <math>\int \frac{-13x-25}{(x+1)^2(x-3)} \cdot dx</math></p> $= \int \frac{3}{(x+1)^2} + \frac{4}{x+1} - \frac{4}{x-3} \cdot dx$ $= \underline{\underline{\frac{-3}{x+1} + 4 \ln x+1  - 4 \ln x-3 }} \quad \underline{1}$ | <p>for each of A, B, C<br/> <math>= 3</math><br/>         Check for carry over error.</p> |

| Suggested solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marker's Feedback |
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| <p>Q.12)(b) <math>\int_1^4 \frac{1}{2+\sqrt{x}} \cdot dx</math></p> <p>Let <math>x = u^2</math>. <math>dx = 2u \cdot du</math></p> <p><math>= \int_{u=1}^{u=2} \frac{1}{2+u} \cdot 2u \cdot du</math> <span style="margin-left: 20px;">1 for change of limits<br/>1 for change of variable<br/>with correct du.</span></p> <p><math>= \int_1^2 \frac{2u+4}{u+2} - \frac{4}{u+2} \cdot du</math> <span style="margin-left: 20px;">1 for this step</span></p> <p><math>= \int_1^2 2 - \frac{4}{u+2} \cdot du</math> <span style="margin-left: 20px;">Note: Can also be done<br/>using <math>x = (u-2)^2</math><br/>etc.<br/><u>4</u></span></p> <p><math>= [2u - 4 \ln u+2 ]_1^2</math></p> <p><math>= \underline{2 - 4 \ln 4 + 4 \ln 3}</math> <span style="margin-left: 20px;">1 for answer.</span></p> <p>Note: Can simplify to <math>2 - 8 \ln 2 + 4 \ln 3</math> etc.<br/>or <math>2 + 4 \ln(\frac{3}{4})</math>.</p> <p>(c) <math>\int [\ln x]^2 \cdot dx</math> <span style="margin-left: 20px;"><math>u = [\ln x]^2</math> <math>v = x</math><br/><math>u' = \frac{2 \ln x}{x}</math> <math>v' = 1</math></span></p> <p>By <math>\int uv' \cdot dx = uv - \int v u' \cdot dx</math></p> <p><math>\int [\ln x]^2 \cdot dx = x [\ln x]^2 - 2 \int \ln x \cdot dx</math> (1) <span style="margin-left: 20px;">1 mark</span></p> <p>Taking <math>\int \ln x \cdot dx</math> out of (1) <math>u = \ln x</math> <math>v = x</math><br/><math>u' = \frac{1}{x}</math> <math>v' = 1</math></p> <p>By <math>\int uv' \cdot dx = uv - \int v u' \cdot dx</math> <span style="margin-left: 20px;"><u>3</u></span></p> <p><math>\int \ln x \cdot dx = x \ln x - \int 1 \cdot dx = \underline{x \ln x - x}</math> (2) <span style="margin-left: 20px;">1 mark</span></p> <p>Sub. (2) in (1):</p> <p><math>\int [\ln x]^2 \cdot dx = x [\ln x]^2 - 2x \ln x + 2x + C</math> <span style="margin-left: 20px;">1 mark</span></p> |                   |

| Suggested solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marker's Feedback |
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| <p>Q. (12)(d) <math>\int \frac{1}{2+\cos x} \cdot dx</math></p> <p>Let <math>t = \tan\left(\frac{x}{2}\right)</math>.</p> $\frac{dt}{dx} = \frac{1}{2} \sec^2\left(\frac{x}{2}\right) = \frac{1+t^2}{2}$ <p><math>\therefore dx = \frac{dx}{dt} \cdot dt = \frac{2}{1+t^2} \cdot dt</math> <i>1 to show <math>dx = \frac{dx}{dt} \cdot dt</math></i></p> $\therefore \int \frac{1}{2+\cos x} \cdot dx$ $= \int \frac{1}{2+\frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} \cdot dt$ $= \int \frac{1}{2+2t^2+1-t^2} \cdot dt$ $= \int \frac{1}{t^2+3} \cdot dt$ <p><i>1 mark</i></p> $= \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{t}{\sqrt{3}}\right) + C.$ $= \frac{1}{\sqrt{3}} \tan^{-1}\left[\frac{\tan\left(\frac{x}{2}\right)}{\sqrt{3}}\right] + C. \quad \text{1 mark}$ <hr/> | <p><u>3</u></p>   |

## Suggested solutions

## Marker's Feedback

Q.(13)(a)(i) Particle is moving with SHM if  $\ddot{x} = -n^2 x$

$$x = 4 \cos \frac{t}{2} - 4\sqrt{3} \sin \frac{t}{2}$$

$$\dot{x} = -2 \sin \frac{t}{2} - 2\sqrt{3} \cos \frac{t}{2}$$

$$\ddot{x} = -\cos\left(\frac{t}{2}\right) + \sqrt{3} \sin \frac{t}{2}$$

$$\ddot{x} = -\frac{1}{4} \left[ 4 \cos \frac{t}{2} - 4\sqrt{3} \sin \frac{t}{2} \right]$$

$$\ddot{x} = -\frac{1}{4} x. \quad \underline{\text{[ to here.]}}$$

Particle is moving with SHM.

$$(ii) 4 \cos \frac{t}{2} - 4\sqrt{3} \sin \frac{t}{2} = A \cos\left(\frac{t}{2} + \alpha\right)$$

$$= A \cos \frac{t}{2} \cos \alpha - A \sin \frac{t}{2} \sin \alpha.$$

Equating parts

$$4 \cos \frac{t}{2} = A \cos \frac{t}{2} \cos \alpha \quad | \quad -4\sqrt{3} \sin \frac{t}{2} = -A \sin \frac{t}{2} \sin \alpha$$

$$A \cos \alpha = 4(1)$$

$$4\sqrt{3} = A \sin \alpha \quad (2)$$

$$(1)^2 + (2)^2: A^2 [\cos^2 \alpha + \sin^2 \alpha] = 4^2 + (4\sqrt{3})^2$$

$$\underline{A = 8}$$

1 for each of  
A &  $\alpha$ .

2

$$\text{Sub. } A = 8 \text{ in (1): } \cos \alpha = \frac{1}{2}$$

$$\text{Sub. } A = 8 \text{ in (2): } \sin \alpha = \frac{\sqrt{3}}{2} \Rightarrow \alpha = \frac{\pi}{3}$$

$$\therefore x = 8 \cos\left(\frac{t}{2} + \frac{\pi}{3}\right)$$

$$\text{Period} = \frac{2\pi}{\left(\frac{1}{2}\right)} = 4\pi \text{ seconds.}$$

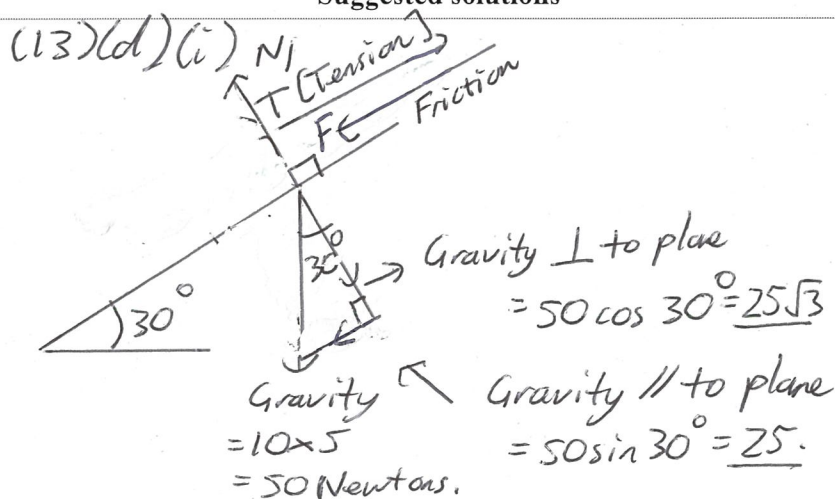
$$\text{Amplitude} = 8 \text{ m.}$$

| Suggested solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marker's Feedback |
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| <p>Q(13)(b) <math>\ddot{x} = -9(x+2)</math></p> <p><math>\frac{d}{dx} \left( \frac{1}{2} v^2 \right) = -9x - 18</math> i to here.</p> <p><math>\frac{d}{dx} (v^2) = -18x - 36.</math></p> <p><math>v^2 = \int -18x - 36 \cdot dx</math></p> <p><math>v^2 = -9x^2 - 36x + C</math></p> <p>As <math> v  = 12 \text{ m/s}</math> when <math>x = 1,</math></p> <p><math>v^2 = 144</math> when <math>x = 1.</math></p> <p><math>144 = -9(1)^2 - 36(1) + C</math></p> <p><math>189 = C</math></p> <p><u><math>v^2 = 189 - 9x^2 - 36x.</math></u>      @</p> <p>By <math>\ddot{x} = -n^2 x, n^2 = 9, n = 3.</math></p> <p><u>Period = <math>\frac{2\pi}{3}</math> seconds.</u>      1 mark</p> <p>Amplitude &amp; centre of motion:</p> <p>Finding where <math>v = 0:</math></p> <p><math>189 - 9x^2 - 36x = 0</math></p> <p><math>x^2 + 4x - 21 = 0</math></p> <p><math>(x+7)(x-3) = 0</math></p> <p><math>x = -7</math> or <math>x = 3.</math></p> <p>Amplitude = 5m. Centre of motion is <u><math>x = -2.</math></u> } 1 mark</p> <p>Note: <math>v^2 = 189 - 9x^2 - 36x</math></p> <p>can also be rearranged as</p> <p><math>v^2 = 9[25 - (x+2)^2]</math></p> <p>From <math>v^2 = n^2(a^2 - (x-C)^2) \rightarrow</math> They CAN do this<br/>this time<br/>(see note below).</p> <p>[Was allowed in 2020 HSC although formula does not appear on reference sheet].</p> |                   |

| Suggested solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marker's Feedback                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Q.(13)(c)</p> <p><math>\rightarrow 16000 \text{ } 10v^2 \leftarrow</math></p> <p>(i) By <math>F = ma</math></p> $1000 \ddot{x} = 16000 - 10v^2 \quad   \text{ mark}$ $\ddot{x} = 16 - 0.01v^2$ <p>Max. <math>v</math> is when <math>\ddot{x} \rightarrow 0</math></p> <p>i.e. <math>16 - 0.01v^2 \rightarrow 0</math></p> $v^2 \rightarrow 1600 \quad \underline{2}$ $v \rightarrow 40 \text{ m/s.} \quad   \text{ mark}$ <p>The car's maximum speed is 40 m/s<br/>[= 144 km/h]</p> <p>(ii) <math>\ddot{x} = v \cdot \frac{dv}{dx} = 16 - 0.01v^2</math></p> $\frac{dv}{dx} = \frac{16 - 0.01v^2}{v}$ $\frac{dx}{dv} = \frac{v}{16 - 0.01v^2}$ $= \frac{100v}{1600 - v^2}$ <p><math>x = -50 \int \frac{-2v}{1600 - v^2} \cdot dv \quad \underline{3}</math></p> $= -50 \ln(1600 - v^2) + C \quad   \text{ mark}$ <p>[note: absolute value not needed as <math>v &lt; 40</math>].</p> <p>As <math>v = 0</math> when <math>x = 0</math>,</p> $0 = -50 \ln 1600 + C \Rightarrow C = 50 \ln 1600$ $x = 50 \ln 1600 - 50 \ln(1600 - v^2)$ $x = 50 \ln \left( \frac{1600}{1600 - v^2} \right) \quad \text{Finding } x \text{ when } v = 30 \text{ km/h}$ $x = 50 \ln \left( \frac{1600}{1600 - 30^2} \right) = \underline{41.3 \text{ m}} \quad   \text{ mark.}$ | <p>Note: As we are finding distance from when <math>v=0</math> to <math>v=v</math>, we can also use</p> $x = -50 \int_0^v \frac{-2v}{1600 - v^2} \cdot dv$ |

## Suggested solutions

## Marker's Feedback



Resolving  $\perp$  to plane:

$$N_1 = 25\sqrt{3}.$$

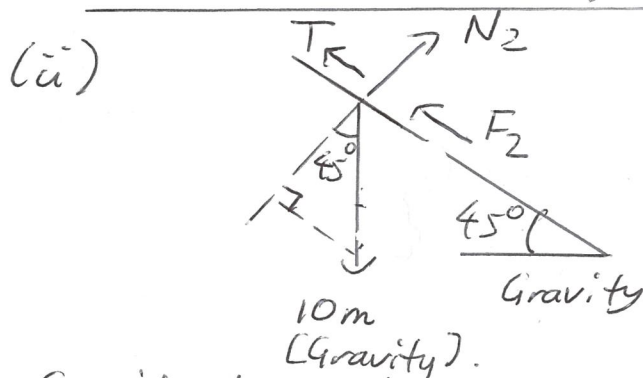
Note:  $F = 0.6N = 15\sqrt{3}.$

Resolving  $\parallel$  to plane:

$$F_1 + 25 = T \quad 1 \text{ mark}$$

$$15\sqrt{3} + 25 = T$$

$$\therefore T = (15\sqrt{3} + 25) \text{ Newtons.}$$



$$\therefore \text{Resolving } \perp \text{ to plane: } N_2 = 10m \cos 45^\circ$$

$$\therefore F_2 = 0.6N_2 = 6m \cos 45^\circ$$

$$= 3m\sqrt{2}. \quad 1 \text{ for } F_2 (1 \text{ mark}).$$

Resolving  $\parallel$  to plane:

$$F_2 + T = 10m \sin 45^\circ$$

$$3m\sqrt{2} + (15\sqrt{3} + 25) = 5m\sqrt{2}$$

$$15\sqrt{3} + 25 = 2m\sqrt{2}$$

$$18.02 = m. \text{ The mass is } 18.02 \text{ kg.}$$

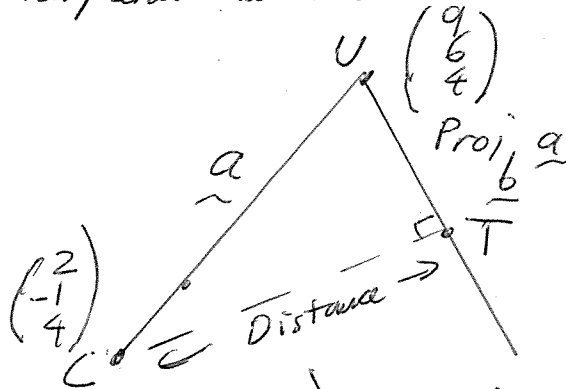
| Suggested solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marker's Feedback |
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| <p>Q. (14)(a) <math>I_N = \int_0^{\frac{\pi}{2}} \sin^N x \cdot dx</math></p> <p>(i) <math>u = \sin^{N-1} x \quad   \quad v = -\cos x</math></p> <p><math>u' = (N-1) \sin^{N-2} x \cos x \quad   \quad v' = \sin x</math></p> <p>By <math>\int u v' \cdot dx = uv - \int v u' \cdot dx</math></p> <p><math>I_N = \left[ -\cos x \sin^{N-1} x \right]_0^{\frac{\pi}{2}} + (N-1) \int_0^{\frac{\pi}{2}} \cos x \sin^{N-2} x \cdot dx</math> <u>make</u></p> <p><math>= 0 + (N-1) \int_0^{\frac{\pi}{2}} \cos^2 x \sin^{N-2} x \cdot dx</math></p> <p><math>(\because \cos \frac{\pi}{2} = \sin 0 = 0)</math> <span style="float: right;"><u><u>2</u></u></span></p> <p><math>I_N = (N-1) \int_0^{\frac{\pi}{2}} \sin^{N-2} x \cdot dx - (N-1) \int_0^{\frac{\pi}{2}} \sin^N x \cdot dx</math> <u>make</u></p> <p>i.e. <math>I_N = (N-1) I_{N-2} - (N-1) I_N</math></p> <p><math>I_N = (N-1) I_{N-2}</math></p> <p><math>I_N = \frac{N-1}{N} I_{N-2}</math></p> <hr/> <p>(ii) <math>I_0 = \int_0^{\frac{\pi}{2}} \sin^0 x \cdot dx = \int_0^{\frac{\pi}{2}} 1 \cdot dx = \frac{\pi}{2}</math> i for <math>I_0</math></p> <p><math>I_2 = \frac{1}{2} I_0 = \frac{\pi}{4}</math></p> <p><math>I_4 = \frac{3}{4} I_2 = \frac{3}{4} \times \frac{\pi}{4} = \frac{3\pi}{16}</math> <span style="float: right;"><u><u>2</u></u></span></p> <p><math>I_6 = \frac{5}{6} I_4 = \frac{5}{6} \times \frac{3\pi}{16} = \frac{5\pi}{32}</math> <u>for <math>I_6</math></u></p> |                   |

## Suggested solutions

## Marker's Feedback

Q. (14)(b) Sphere: Centre =  $\begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$   $r = \sqrt{14}$  units.  $\underline{2}$   
 (i)  $\text{unit}$   $\text{unit}$

(ii) Perpendicular distance:



$$\underline{a} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} - \begin{pmatrix} 9 \\ 6 \\ 4 \end{pmatrix} = \begin{pmatrix} -7 \\ -7 \\ 0 \end{pmatrix}$$

$\underline{b}$  = direction vector of line =  $\begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix}$

Perpendicular distance =  $CT$

$$= |\underline{a} - \text{Proj}_{\underline{b}} \underline{a}|$$

$$\text{Noting } \text{Proj}_{\underline{b}} \underline{a} = \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|^2} \underline{b}$$

$$\begin{aligned} \text{Proj}_{\underline{b}} \underline{a} &= \frac{\begin{pmatrix} -7 \\ -7 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix}}{4^2 + 5^2 + 1^2} \begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix} \\ &= -\frac{3}{2} \begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix} = \begin{pmatrix} -6 \\ -7\frac{1}{2} \\ -\frac{3}{2} \end{pmatrix} \text{ i for } \text{Proj}_{\underline{b}} \underline{a} \end{aligned}$$

$$\begin{aligned} \therefore \underline{a} - \text{Proj}_{\underline{b}} \underline{a} &= \begin{pmatrix} -7 \\ -7 \\ 0 \end{pmatrix} - \begin{pmatrix} -6 \\ -7\frac{1}{2} \\ -\frac{3}{2} \end{pmatrix} \\ &= \begin{pmatrix} -1 \\ -\frac{1}{2} \\ \frac{1}{2} \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \& \perp \text{ distance} &= |\underline{a} - \text{Proj}_{\underline{b}} \underline{a}| = \sqrt{(-1)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} \\ &= \sqrt{3\frac{1}{2}} \\ &\doteq \underline{1.87 \text{ units.}} \end{aligned}$$

## Suggested solutions

## Marker's Feedback

(14)(b) Note that as the perpendicular distance (u) is less than the sphere's radius, (continued) the line must pass THROUGH the sphere, therefore hitting it in 2 places.

for answer  
= 1.87  
with this explanation.

Note: ALTERNATIVE PERPENDICULAR DISTANCE METHOD (was asked in guided fashion in 2024 NEISA paper).

(OR)

All points on the line L have equation

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 9+4\lambda \\ 6+5\lambda \\ 4+\lambda \end{pmatrix} \text{ \& so the vector from}$$

C to any point on L is given by

$$\begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} - \begin{pmatrix} 9+4\lambda \\ 6+5\lambda \\ 4+\lambda \end{pmatrix} = \begin{pmatrix} -7-4\lambda \\ -7-5\lambda \\ -\lambda \end{pmatrix}$$

\& the magnitude of this vector

$$= \sqrt{(-7-4\lambda)^2 + (-7-5\lambda)^2 + (-\lambda)^2}$$

$$\therefore (\text{Distance from C to L})^2$$

$$= (-7-4\lambda)^2 + (-7-5\lambda)^2 + (-\lambda)^2$$

$$= 42\lambda^2 + 126\lambda + 98$$

i for this expression

Finding the value of  $\lambda$  for minimum

$$\text{distance, } \lambda = \frac{-126}{84} = -\frac{3}{2}$$

$$\& (\text{minimum distance})^2 = 42\left(-\frac{3}{2}\right)^2 + 126\left(-\frac{3}{2}\right) + 98$$

$$= \underline{3.5}$$

$$\text{So perpendicular distance} = \sqrt{3.5} \quad \& \text{ for } \sqrt{3.5}$$

$$= \underline{1.87 \text{ units}}$$

\& why it's  
inside  
circle.

| Suggested solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marker's Feedback                   |
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| <p>(14)(b)(iii) Points of intersection of line &amp; sphere;</p> <p>Line <math>L</math>: <math>\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 9 + 4\lambda \\ 6 + 5\lambda \\ 4 + \lambda \end{pmatrix}</math></p> <p>Sub. into <math>(x-2)^2 + (y+1)^2 + (z-4)^2 = 14</math>.</p> $(7+4\lambda)^2 + (7+5\lambda)^2 + (\lambda)^2 = 14. \quad 1 \text{ mark.}$ $42\lambda^2 + 126\lambda + 98 = 14$ $42\lambda^2 + 126\lambda + 84 = 0$ $\lambda^2 + 3\lambda + 2 = 0$ $(\lambda+1)(\lambda+2) = 0$ $\lambda = -1 \text{ or } \lambda = -2. \quad 1 \text{ mark.}$ <p>If <math>\lambda = -1</math>, <math>\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \\ 3 \end{pmatrix}</math></p> <p>If <math>\lambda = -2</math>, <math>\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ -4 \\ 2 \end{pmatrix}</math> } 1 mark</p> | <p style="text-align: right;">3</p> |
| <p>(iv) <math>L_1 \perp L_2</math> if dot product of direction vectors = 0.</p> $\begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} = -8 + 5 + 3 = 0$ <p><u><math>L_1 \perp L_2</math></u> 1 mark.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                     |

| Suggested solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marker's Feedback                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <p>(14)(b)(v) Intersection of <math>L_1</math> &amp; <math>L_2</math>:</p> $\begin{pmatrix} 9 \\ 6 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}$ <p> <math>x: 9 + 4\lambda = 2 - 2\mu \Rightarrow 4\lambda + 2\mu = -7 \quad (1)</math><br/> <math>y: 6 + 5\lambda = -1 + \mu \Rightarrow 5\lambda - \mu = -7 \quad (2)</math><br/> <math>z: 4 + \lambda = 4 + 3\mu \Rightarrow \lambda - 3\mu = 0</math><br/> <math>\quad \quad \quad \text{or } \lambda = 3\mu \quad (3)</math> </p> <p>Sub. (3) in (1): <math>12\mu + 2\mu = -7 \Rightarrow \mu = -\frac{1}{2}</math>.</p> <p>From (3), as <math>\lambda = 3\mu</math>, <math>\lambda = -\frac{3}{2}</math>.</p> <p>Check to see if <math>\lambda = -\frac{3}{2}</math>, <math>\mu = -\frac{1}{2}</math> works in (2):</p> <p> LHS: <math>6 + 5 \times -\frac{3}{2} = -\frac{3}{2}</math><br/> RHS: <math>-1 - \frac{1}{2} = -\frac{3}{2}</math><br/> <math>\lambda = -\frac{3}{2}</math>, <math>\mu = -\frac{1}{2}</math> works in (2). </p> <p><math>\therefore</math> Point of intersection</p> $= \begin{pmatrix} 9 \\ 6 \\ 4 \end{pmatrix} - \frac{3}{2} \begin{pmatrix} 4 \\ 5 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} 3 \\ -\frac{3}{2} \\ \frac{5}{2} \end{pmatrix}$ <p>or</p> $= \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}$ $= \begin{pmatrix} 3 \\ -\frac{3}{2} \\ \frac{5}{2} \end{pmatrix}$ | <p>1 mark.</p> <p>1 for testing in unused equation (in case skew).</p> <p>3</p> <p>1 for answer</p> |

| Suggested solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marker's Feedback |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <p>Q. (15)(a)</p> <p>Show true for <math>n =</math></p> $\begin{array}{ll} \text{LHS} & \text{RHS} \\ = 7! & = 3^7 \\ = 5040 & = 2187 \end{array}$ <p style="text-align: center;"><math>\text{LHS} &gt; \text{RHS}</math></p> <p>True for <math>n = 7</math>.</p> <p>Assume true for <math>n = k, k \geq 7</math>.<br/>i.e. <math>k! &gt; 3^k</math>.</p> <p>Prove true for <math>n = k+1, k \geq 7</math>.<br/>i.e. RTP <math>(k+1)! &gt; 3^{k+1}</math>.</p> $\begin{array}{ll} \text{LHS} & 3 \\ = (k+1)! & \\ = (k+1) \times k! & \\ > (k+1) \times 3^k \text{ [by assumption]} & \text{1 for using assumption} \\ > 3 \times 3^k \text{ [as } k \geq 7, k+1 > 3 \text{]} & \text{1 for } k+1 > 3. \\ = 3^{k+1} & \\ \therefore (k+1)! > (k+1) \times 3^k > 3^{k+1} & \\ \underline{(k+1)! > 3^{k+1}} & \end{array}$ <p>If it is true for <math>n = k</math> it will be true for <math>n = k+1</math>.</p> <p>Hence as it is true for <math>n = 7</math> it will be true <math>\forall n \in \mathbb{Z}^+ \geq 7</math> by the principle of mathematical induction.</p> |                   |

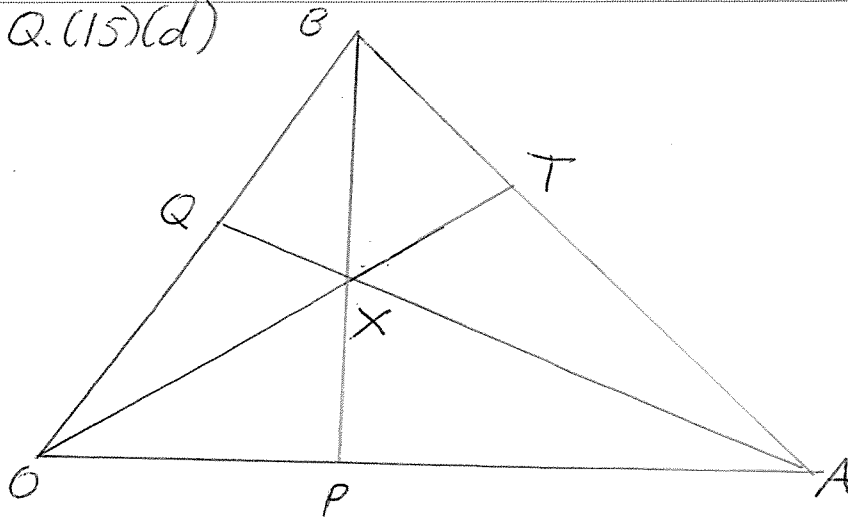
| Suggested solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marker's Feedback |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <p>Q. (15)(b)(i) If <math>a, b</math> real<br/> <math>(a-b)^2 \geq 0</math> [as <math>a-b</math> real].<br/> <math>\therefore a^2 - 2ab + b^2 \geq 0</math><br/> <math>\underline{a^2 + b^2 \geq 2ab.}</math> <u>1 mark.</u></p> <p>(ii) Hence for <math>a, b, c</math> real<br/> <math>a^2 + b^2 \geq 2ab</math> (1)<br/> <math>a^2 + c^2 \geq 2ac</math> (2)<br/> <math>b^2 + c^2 \geq 2bc</math> (3) <u>1 mark</u><br/> <math>(1) + (2) + (3)</math><br/> <math>2(a^2 + b^2 + c^2) \geq 2(ab + ac + bc)</math><br/> <math>\underline{a^2 + b^2 + c^2 \geq ab + ac + bc.}</math></p> <p>(iii) Using (ii)<br/> <math>\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \geq \frac{1}{ab} + \frac{1}{ac} + \frac{1}{bc}</math><br/> <math>= \frac{c}{abc} + \frac{b}{abc} + \frac{a}{abc}</math><br/> <math>= \frac{a + b + c}{abc.}</math> <u>1 mark.</u></p> <p><math>\therefore \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \geq \frac{a+b+c}{abc.}</math></p> |                   |

| Suggested solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marker's Feedback |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <p>Q. (15)(c) Let <math>\sqrt{29} = \frac{p}{q}</math>, <math>p, q \in \mathbb{Z}^+</math><br/>         &amp; <math>p</math> &amp; <math>q</math> have no common factors.<br/> <math>\therefore \frac{p^2}{q^2} = 29</math><br/> <math>p^2 = 29q^2 \quad (1)</math><br/>         As <math>p, q</math> have no common factors &amp;<br/> <math>29</math> is prime, as <math>29</math> is a factor of <math>p^2</math><br/> <math>29</math> is a factor of <math>p</math>. <math>\rightarrow</math> 1 mark to here.<br/> <math>\therefore p = 29k</math>, <math>k \in \mathbb{Z}^+</math><br/> <math>p^2 = 841k^2 \quad (2)</math><br/>         Sub. (2) in (1):<br/> <math>841k^2 = 29q^2</math><br/> <math>29k^2 = q^2</math><br/>         As <math>29</math> is a factor of <math>q^2</math>, <math>29</math> is a factor<br/>         of <math>q</math> (as <math>29</math> is prime).<br/> <math>\therefore 29</math> is a factor of both <math>p</math> &amp; <math>q</math>. <math>\rightarrow</math> 1 mark to here.<br/>         But <math>p</math> &amp; <math>q</math> have no common factors.<br/> <math>\therefore</math> By contradiction, <math>\sqrt{29}</math> must be<br/>         irrational.</p> |                   |

## Suggested solutions

## Marker's Feedback

Q. (15)(d)



$$(i) \vec{OX} = \vec{OP} + \vec{PX} \\ = \frac{1}{3} \underline{a} + \lambda \vec{PB} \quad 1 \text{ mark}$$

$$\text{Noting } \vec{PB} = \vec{PO} + \vec{OB} = -\frac{1}{3} \underline{a} + \underline{b} \quad (2) \quad 2$$

$$\vec{OX} = \frac{1}{3} \underline{a} + \lambda \left( -\frac{1}{3} \underline{a} + \underline{b} \right) \quad 1 \text{ mark}$$

$$\vec{OX} = \frac{1}{3} (1 - \lambda) \underline{a} + \lambda \underline{b} \text{ as required.}$$

$$(ii) \text{ As } \vec{OX} \text{ also } = \mu \underline{a} + \frac{1}{2} (1 - \mu) \underline{b}$$

Equating (i) &amp; (ii)

$$\underline{a} : \frac{1}{3} (1 - \lambda) = \mu$$

$$\underline{2 - 2\lambda} = 6\mu \quad (1)$$

$$\underline{b} : \frac{1}{2} = \frac{1}{2} (1 - \mu) \quad (2)$$

$$2\lambda = 1 - \mu \quad (2)$$

1 mark for the 2 equations

Sub. (2) in (1):

$$2 - (1 - \mu) = 6\mu$$

$$\underline{\mu = \frac{1}{5}}$$

Sub.  $\mu = \frac{1}{5}$  in (1)

$$2 - 2\lambda = \frac{6}{5}$$

$$\underline{\lambda = \frac{2}{5}}$$

1 mark for  $\lambda$  &  $\mu$ 

$$\lambda = \frac{2}{5}, \mu = \frac{1}{5}$$

$$\vec{OX} = \frac{1}{3} \left( 1 - \frac{2}{5} \right) \underline{a} + \frac{2}{5} \underline{b}$$

[using (i)]

$$\vec{OX} = \frac{1}{5} \underline{a} + \frac{2}{5} \underline{b}$$

3  
1 mark  
for finish

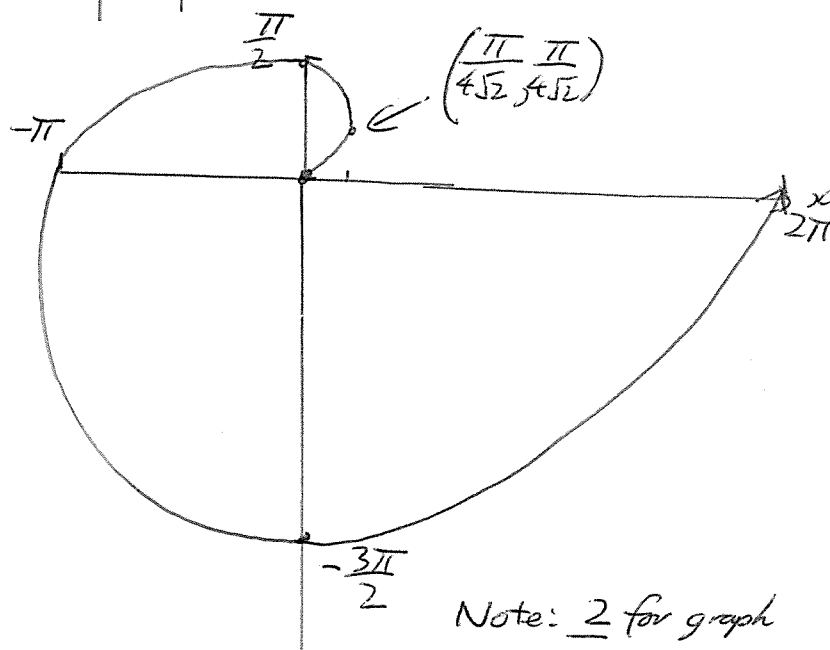
| Suggested solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marker's Feedback                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <p>(d) Q.15) (b) (iii) if <math>T</math> is collinear with <math>O</math> &amp; <math>X</math><br/> then <math>\vec{OT} = k \vec{OX}</math><br/> <math display="block">\vec{OT} = k \left( \frac{1}{5} \underline{a} + \frac{2}{5} \underline{b} \right) \quad (1)</math> <math display="block">\begin{aligned} \vec{OT} \text{ also} &amp;= \vec{OB} + \vec{BT} \\ &amp;= \underline{b} + h \vec{BA} \\ &amp;= \underline{b} + h (\underline{a} - \underline{b}) \\ &amp;= h \underline{a} + (1-h) \underline{b} \quad (2) \end{aligned}</math> <p>Equating parts in (1) &amp; (2)</p> <math display="block">\underline{a}: \frac{1}{5} k \underline{a} = h \underline{a} \Rightarrow \frac{1}{5} k = h \quad (3)</math> <math display="block">\underline{b}: \frac{2}{5} k \underline{b} = (1-h) \underline{b} \Rightarrow \frac{2}{5} k = 1-h \quad (4)</math> <p>(3) + (4): <math>\frac{3}{5} k = 1</math><br/> <math>k = \frac{5}{3}</math></p> <p>∴ Using (1):</p> <math display="block">\begin{aligned} \vec{OT} &amp;= \frac{5}{3} \left( \frac{1}{5} \underline{a} + \frac{2}{5} \underline{b} \right) \\ &amp;= \underline{\underline{\frac{1}{3} \underline{a} + \frac{2}{3} \underline{b}}} \end{aligned}</math> <p>[Note: Check: Can also use<br/> <math>\vec{OT} = h \underline{a} + (1-h) \underline{b}</math> As <math>k = \frac{5}{3}</math>, <math>h = \frac{1}{5} k = \frac{1}{3}</math>.<br/> <math>= \underline{\underline{\frac{1}{3} \underline{a} + \frac{2}{3} \underline{b}}}</math></p> </p> | <p>1 mark for<br/>2 equations</p> <p>1 mark for<br/>equating<br/><math>\underline{a}</math> &amp; <math>\underline{b}</math> parts.</p> |

## Suggested solutions

## Marker's Feedback

Q. (16)(a)

|     |   |                         |                 |        |                   |        |
|-----|---|-------------------------|-----------------|--------|-------------------|--------|
| $t$ | 0 | $\frac{\pi}{4}$         | $\frac{\pi}{2}$ | $\pi$  | $\frac{3\pi}{2}$  | $2\pi$ |
| $x$ | 0 | $\frac{\pi}{4\sqrt{2}}$ | 0               | $-\pi$ | 0                 | $2\pi$ |
| $y$ | 0 | $\frac{\pi}{4\sqrt{2}}$ | $\frac{\pi}{2}$ | 0      | $-\frac{3\pi}{2}$ | 0      |



with  $t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$ .

[ $t = \frac{\pi}{4}$  was just a check for the spiral shape].

1 for table without graph.

| Suggested solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marker's Feedback |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <p>Q(16)(b)(i) Solutions to <math>z^9 - 1 = 0</math> are</p> <p><math>z = \text{cis } \frac{2k\pi}{9}</math>, <math>k = 0, 1, 2, 3, 4, 5, 6, 7, 8</math>. <math>\rightarrow</math> Satisfactory <u>1</u><br/>to list like this</p> <p><math>\therefore z = \text{cis } \frac{2\pi}{9}, \text{cis } \frac{4\pi}{9}, \text{cis } \frac{2\pi}{3}, \text{cis } \frac{8\pi}{9}, \text{cis } \frac{10\pi}{9},</math> <u>or</u><br/><math>\text{cis } \frac{4\pi}{3}, \text{cis } \frac{14\pi}{9}, \text{cis } \frac{16\pi}{9}, 1.</math> using principal argument.</p> <p>(ii) As <math>z^9 - 1 = (z^3 - 1)(z^6 + z^3 + 1)</math><br/>&amp; the solutions to <math>z^3 - 1 = 0</math> are<br/><math>\text{cis } \frac{2\pi}{3}, \text{cis } \frac{4\pi}{3}</math> &amp; 1.</p> <p>the solutions to <math>z^6 + z^3 + 1 = 0</math> are<br/><math>z = \text{cis } \frac{2\pi}{9}, \text{cis } \frac{4\pi}{9}, \text{cis } \frac{8\pi}{9},</math> <u>1</u><br/><math>\text{cis } \frac{10\pi}{9}, \text{cis } \frac{14\pi}{9}, \text{cis } \frac{16\pi}{9}.</math></p> <p>(iii) If <math>z = \cos \theta + i \sin \theta</math>,<br/> <math>z^n = \cos n\theta + i \sin n\theta</math> (by De Moivre).<br/> <math>z^{-n} = \cos(-n\theta) + i \sin(-n\theta)</math><br/> <math>= \cos n\theta - i \sin n\theta</math> (as cos even, sin odd) } Need both for <u>1</u></p> <p>Hence, <math>z^n + z^{-n}</math><br/> <math>= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta</math><br/> <math>= \underline{2 \cos n\theta}.</math></p> |                   |

## Suggested solutions

## Marker's Feedback

$$Q.16(b)(iv) \quad z^6 + z^3 + 1$$

$$= (z - \text{cis } \frac{2\pi}{9})(z - \text{cis } \frac{16\pi}{9})(z - \text{cis } \frac{4\pi}{9})(z - \text{cis } \frac{14\pi}{9})$$

$$(z - \text{cis } \frac{8\pi}{9})(z - \text{cis } \frac{10\pi}{9}) \quad \underline{1 \text{ mark}}$$

$$\text{Noting } (z - \text{cis } \frac{2\pi}{9})(z - \text{cis } \frac{16\pi}{9})$$

$$= (z - \text{cis } \frac{2\pi}{9})(z - \text{cis } (-\frac{2\pi}{9}))$$

$$= (z^2 - z(\text{cis } \frac{2\pi}{9} + \text{cis } -\frac{2\pi}{9}) + \text{cis } \frac{2\pi}{9} \times \text{cis } (-\frac{2\pi}{9}))$$

$$= (z^2 - 2z \cos \frac{2\pi}{9} + 1) \quad \text{[using (iii)]} \quad \underline{1}$$

Similarly,

Total to be 2 marks to be.

$$(z - \text{cis } \frac{4\pi}{9})(z - \text{cis } \frac{14\pi}{9}) = z^2 - 2z \cos \frac{4\pi}{9} + 1.$$

$$(z - \text{cis } \frac{8\pi}{9})(z - \text{cis } \frac{10\pi}{9}) = z^2 - 2z \cos \frac{8\pi}{9} + 1.$$

Hence

$$z^6 + z^3 + 1$$

$$= (z^2 - 2z \cos \frac{2\pi}{9} + 1)(z^2 - 2z \cos \frac{4\pi}{9} + 1)$$

$$(z^2 - 2z \cos \frac{8\pi}{9} + 1)$$

(v) Dividing (iv) by  $z^3$ :

$$\frac{(z^6 + z^3 + 1)}{z^3} = \left( \frac{z^2 - 2z \cos \frac{2\pi}{9} + 1}{z} \right) \left( \frac{z^2 - 2z \cos \frac{4\pi}{9} + 1}{z} \right) \left( \frac{z^2 - 2z \cos \frac{8\pi}{9} + 1}{z} \right)$$

$$z^3 + \frac{1}{z^3} + 1 = \left( z + \frac{1}{z} - 2 \cos \frac{2\pi}{9} \right) \left( z + \frac{1}{z} - 2 \cos \frac{4\pi}{9} \right) \left( z + \frac{1}{z} - 2 \cos \frac{8\pi}{9} \right) \quad \text{1 to be.}$$

$$\text{As } z^n + \frac{1}{z^n} = 2 \cos n\theta.$$

$$2 \cos 3\theta + 1 = (2 \cos \theta - 2 \cos \frac{2\pi}{9})(2 \cos \theta - 2 \cos \frac{4\pi}{9})(2 \cos \theta - 2 \cos \frac{8\pi}{9}) \quad \underline{2}$$

$$2 \cos 3\theta + 1 = 8 (\cos \theta - \cos \frac{2\pi}{9})(\cos \theta - \cos \frac{4\pi}{9})(\cos \theta - \cos \frac{8\pi}{9}) \quad \underline{2}$$

$$\cos 3\theta + \frac{1}{2} = 4 (\cos \theta - \cos \frac{2\pi}{9})(\cos \theta - \cos \frac{4\pi}{9})(\cos \theta + \cos \frac{\pi}{9}) \quad \text{As } \cos \frac{8\pi}{9} = -\cos \frac{\pi}{9}.$$

as required.

2

## Suggested solutions

## Marker's Feedback

Q.(16)(c) [Note: Two alternative solutions:  
1<sup>st</sup> if done in 1 hit, 2<sup>nd</sup> if restarting  
the question when the projectile starts to fall]

1<sup>st</sup> way:  $\downarrow$   $\downarrow$   
Gravity: 10m 0.05

By  $F = ma$  [with up as positive],

$$m\ddot{x} = -10m - 0.05v$$

$$\ddot{x} = -10 - 0.05v$$

1

[Note: On the way down,  $v$  is negative

$\therefore -0.05v$  would be pushing UP so

$\ddot{x} = -10 - 0.05v$  would still be  
valid on the way down].

$$\therefore \frac{dv}{dt} = -10 - 0.05v$$

$$\frac{dt}{dv} = \frac{-1}{10 + 0.05v}$$

$$t = \int \frac{-20}{200 + v} \cdot dv$$

$$= -20 \ln(200 + v) + C$$

When  $t = 0$ ,  $v = v_0$ : (\*)

$$\therefore 0 = -20 \ln(200 + v_0) + C$$

$$C = 20 \ln(200 + v_0)$$

$$\therefore t = -20 \ln\left(\frac{200 + v}{200 + v_0}\right)$$

1

$$e^{-\frac{t}{20}} = \frac{200 + v}{200 + v_0}$$

$$(200 + v_0)e^{-\frac{t}{20}} - 200 = v$$

(\*) Note: Could also use a definite  
integral & do  $\int_{v_0}^v \frac{-20}{200 + v} \cdot dv$

1

| Suggested solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marker's Feedback |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| $x = \int (200 + v_0) e^{-\frac{t}{20}} - 200 \cdot dt$ $= -(4000 + 20v_0) e^{-\frac{t}{20}} - 200t + C$ <p>As <math>x = 0</math> when <math>t = 0</math>,</p> $0 = -(4000 + 20v_0) e^0 + C$ $x = -(4000 + 20v_0) e^{-\frac{t}{20}} - 200t + (4000 + 20v_0) \cdot \frac{1}{20}$ <p>As projectile is at <math>x = 0</math> when <math>t = 5</math>,</p> $0 = -(4000 + 20v_0) e^{-\frac{1}{4}} - 1000 + (4000 + 20v_0) \cdot \frac{1}{20}$ $4000e^{-\frac{1}{4}} - 3000 = 20v_0(1 - e^{-\frac{1}{4}})$ $26.04 \approx v_0$ <p>Initial velocity is <u>26.04 m/s.</u> <u>ANSWER.</u> <u>1</u></p> |                   |

## Suggested solutions

## Marker's Feedback

Alternative Solution from starting again at the top:

[Noting 3 to here already = marks].

1<sup>st</sup> find maximum height: is  $x$  when  $v=0$

$x$  when going up was

$$-(4000 + 20v_0)e^{-\frac{t}{20}} - 200t + (4000 + 20v_0)$$

Time when  $v=0$  was

$$t = -20 \ln\left(\frac{200}{200+v_0}\right) \Rightarrow e^{-\frac{t}{20}} = \frac{200}{200+v_0}$$

$$\text{So } x = -(4000 + 20v_0) \times \frac{200}{200+v_0}$$

$$-200x - 20 \ln\left(\frac{200}{200+v_0}\right) + (4000 + 20v_0)$$

$$x = 4000 \ln\left(\frac{200}{200+v_0}\right) + 20v_0$$

Now starting at top: down positive.

$$t=0.$$

$$\downarrow \quad \ddot{x} = 10 - 0.05v$$

$$0.05mv \frac{dv}{dt} = 10 - 0.05v$$

$$\uparrow \quad \frac{dt}{dv} = \frac{1}{10 - 0.05v}$$

$$= \frac{20}{200 - v}$$

$$t = -20 \int \frac{1}{200-v} dv \quad \left\{ \begin{array}{l} \text{Don't need absolute value} \\ \text{as } v < 200. \end{array} \right.$$

$$= -20 \ln(200-v) + C$$

As  $v=0$  when  $t=0$  [starting at top].

$$= -20 \ln(200) + C \Rightarrow C = 20 \ln 200$$

$$t = -20 \ln\left(\frac{200-v}{200}\right)$$

1 mark.

## Suggested solutions

## Marker's Feedback

Alternative Solution Starting Again at Top:  
(continued).

$$e^{-\frac{t}{20}} = \frac{200-v}{200}$$

$$v = 200 - 200e^{-\frac{t}{20}}$$

$$x = \int 200 - 200e^{-\frac{t}{20}} dt$$

$$x = 200t + 4000e^{-\frac{t}{20}} + C$$

As  $x=0$  when  $t=0$

$$0 = 4000 + C \Rightarrow C = -4000$$

$$x = 200t + 4000e^{-\frac{t}{20}} - 4000.$$

As projectile takes 5 seconds to land,  
it will fall from top height of

$$4000 \ln\left(\frac{200}{200+v_0}\right) + 20v_0$$

in  $5 + 20 \ln\left(\frac{200}{200+v_0}\right)$  seconds [5 seconds - time to top].

$$\therefore \text{By } x = 200t + 4000e^{-\frac{t}{20}} - 4000$$

$$\left. \begin{aligned} 4000 \ln\left(\frac{200}{200+v_0}\right) + 20v_0 &= 200\left(5 + 20 \ln\left(\frac{200}{200+v_0}\right)\right) \\ + 4000 \times \frac{e^{-\frac{1}{4}(200+v_0)}}{200} &= 4000 \end{aligned} \right\} \text{1 mark}$$

$$\left[ \text{Note: If } t = 5 + 20 \ln\left(\frac{200}{200+v_0}\right), e^{-\frac{t}{20}} = e^{-\frac{1}{4} \ln\left(\frac{200+v_0}{200}\right)} = e^{-\frac{1}{4}(200+v_0)} \right]$$

$$20v_0 = 20e^{-\frac{1}{4}(200+v_0)} - 3000$$

$$20v_0(1 - e^{-\frac{1}{4}}) = 4000e^{-\frac{1}{4}} - 3000$$

$$v_0 = \frac{4000e^{-\frac{1}{4}} - 3000}{2(1 - e^{-\frac{1}{4}})}$$

$$= 26.04 \text{ m/s as required. } \text{1 mark}$$

END OF SOLUTIONS!!!